

Pinghe Jinding Library Consulting Team Recruiting Notice

Task Introduction

The library of Pinghe School Jinding Campus is a very popular place. With a rich collection of books and a pleasant environment, it is a good place for students to study and read. The librarians have always been committed to providing students with better services, and are therefore openly recruiting consulting experts. Your group has decided to apply. To do so, you need to submit a report and give a 15-minute presentation to demonstrate your analysis of the circulation records, and based on that, provide an analysis of the library's current operations and suggestions for the future, as well as a recommended book purchase list.

You will be given a dataset of circulation records for IB students of the Jinding Library for the 2024-2025 academic year, including book information, borrowing/return times, students' grades, etc. The detailed data will be provided separately in Excel format. Specific descriptions of data fields are provided at the end of this document.

What to do?

Interesting Fact

1. Organize the information and data chosen from the given data set. Measuring the center and spread or other aspect of data on several individual data fields. Present several statistical characteristics, and use appropriate forms, such as tables, diagrams and/or charts, to display the information. For example, using the title field, you can calculate how many different books have been borrowed, which book is the most popular, the distribution of borrowing counts for all books, etc. Using circulation time, you can calculate on which dates borrowing was highest/lowest, distribution across

months, etc.

2. For the data in Task 1, further perform statistics broken down by grade and make comparisons. Note that Class 22 has only 8 classes, while Class of 23 and 24 each have 10 classes.

The above two tasks should focus only on "Borrow" and "Renew" data. Moreover, every member of the group must independently analyze at least one individual data field and use at least one chart to complete Tasks 1 & 2. Choices among group members must not overlap, and creativity and imagination are encouraged.

Bivariate Statistics

3. Borrowing duration or borrowing frequency: First, match borrowing and return information for the same book in the dataset to obtain the duration for which that book was borrowed (recommended unit: "days"). For records that have only borrow or only return, you may delete them, or assume that the book was borrowed on the first day of this school year or returned on the last day of this school year. (Regardless of the method used, please clearly state your approach and assumptions.) Analyze the correlation between some (at least two) data field in the dataset and borrowing duration. You may analyze the whole student population, or separately by grade. You may also calculate the borrowing frequency of the same book and analyze the relationship between some data field and borrowing frequency.

In the first three tasks, only data provided by the library may be used. Please note that there may be multiple copies of the same book in the library, each with a different collection barcode number for identification. Therefore, when calculating borrowing duration or borrowing frequency, you should distinguish between different copies of the same book, and then you can combine them to calculate the total borrowing duration and total borrowing frequency for that type of book.

Collect New Data

4. Consider **explanatory** factors (**categorical or quantitative**) that **may** have a **causal** relationship with borrowing duration or borrowing frequency, and collect new data using the following two methods:
- a) Search in public information on the Internet (websites or apps) such as book ratings, number of reviews, genre information, etc.
 - b) Design a questionnaire, but **limited to this class only**. To ensure fairness of the recruitment and avoid cross-group copying, strictly distribute the questionnaire **between 48 and 49 hours after you receive this recruitment notice** – do not distribute earlier or later. Besides collect data, the survey may also ask students for their suggestions about the library, or other aspects you are interested in.
5. Organize the collected data and use charts to present and analyze them. Discuss the following regarding the considered factors:
- a) Value range or distribution characteristics
 - b) Correlation and causality with borrowing frequency and borrowing duration
 - c) Other aspects worth discussing

Analysis and Improvement Plan

6. Based on all the above results obtained in tasks 1-5, analyze the current state of library operations and provide recommendations. You can conduct more original data analysis and provide better recommendations based on that. These can be considered as optional tasks (see later in the document). Again, creativity and imagination are encouraged.
7. Design an improvement plan for the library for the 2026-27 academic year with a budget not exceeding RMB 10,000. This plan should include at least one book acquisition plan, including information on the books, the quantity to be purchased, and the reasons for recommendation. The books to be purchased are mainly new books, but it's also possible to buy a few more of the books already in the library.

The remaining budget can be used to improve library facilities and services. All expenditures must be feasible and have price references. If volunteer services are involved, please note that this approach cannot involve unlimited recruitment and will require certain management costs (such as the effort required for recruitment, training, and management) and personnel costs (such as volunteer certificates, thank-you gifts, and a small amount of food and transportation allowances).

Optional Tasks

The following are optional tasks:

- Predict borrowing of IB students in the 25-26 academic year.
- Predict borrowing of the books you recommend in the 26-27 academic year.
- Other data analysis you think can be done. (As mentioned before in Task 6)

About Submission

As a result, your group is supposed to :

- i. submit an organized, complete and concise report
 - a) at the beginning, indicate the dataset used (see details below)
 - b) with 12-20 pages of main body :
 - c) including the improvement plan
 - d) with the survey (if any) as appendix
 - e) with brief description of your use of AI in the appendix (Details will follow)
 - f) at the very end, use a table to indicate the division of work among group members (see below)
- ii. prepare a presentation within 15 minutes

Division of Work	Name1	Name2	Name3	Name4
Data cleaning / Data processing				
Interesting Facts (should involve	√	√	√	√

everyone)				
Analysis on factors related to borrowing duration / frequency				
Questionnaire design / analysis				
Improvement Plan Design				
Library operation analysis				
Presentation File Preparation				
.....				

- The content of the first column can be defined by each group
- If a certain task is primarily done by one person with contributions from others, you may use two √ to indicate dominance, and one √ for participation.

To the recruiter: When writing the report, you should avoid the question-answer format. The above “what to do” are some recommended guidance, you can make appropriate additions and amendments based on that. Please also remember to add a list of reference for the borrowed ideas, information and/or data source.

Data field descriptions

Collection Barcode: If there are multiple copies of the same book, each copy has a different barcode. When analyzing borrowing duration, make sure to match borrowing and return information for the same barcode.

Title: The title of the book.

ISBN: The International Standard Book Number is a numeric commercial book identifier that is intended to be unique.

Circulation Type: Includes "Borrow" (外借), "Renew" (续借), "Return" (归还). You may delete renewals or treat them as a new borrow, depending on your group's interpretation of the data.

Circulation Time: Accurate to the second. Therefore, you can perform statistical analysis on month, day, even morning/afternoon, or specific hour or even seconds just for fun.

Price, Author, Publisher, Publication time, Place of Publication, Pages: Note that these information are incomplete and the formatting is inconsistent. If you want to analyze them, manual data cleaning (or using AI assistance) is required.

Classification number: Those starting with “IB” refer to IB textbooks, a self-created classification used by Jinding Library. Those starting with other letters are based on the Chinese Library Classification(《中国图书馆分类法》) for Chinese books in the library. Those starting with numbers are based on the *Dewey Decimal Classification and Relative Index* for English books in the library. You may consult these standard documents listed in the References part of this document to understand and use this data field. You may also use other methods for book classification analysis.

UserID: Each student has a unique UserID, which does not directly correspond to their student ID. Therefore, the whole dataset can be considered anonymised. You can use it for analysis, e.g., the number of books borrowed by the most active borrower, etc.

Grade: In this dataset, we distinguish students of different grades by their year of enrollment. For simplicity, IB22G12 represents IB students who enrolled in 2022 and are in grade 12 in the year the dataset is set, and so on.

Monthly assessment instructions

- **This monthly assessment is a collaboration with the library. Outstanding reports and presentations may be displayed in the library or on the official public account.**
- **Submission deadlines:** Submit report by **May 16, 23:59**; submit presentation file by **May 17, 23:59**. For in-class presentations, the students will use the submitted file as in teachers' hand.
- **Every member must submit a report.** Reports within the same group should be

identical, which signifies your endorsement of every word in the report, including the division of work.

- **Report length and font requirements**

- a) Similar to IA : **main body 12 – 20 pages.**

- b) Should match the 15-minute presentation. Not too long or too short. Do not blindly stuff words; charts should not be too large. More charts will naturally increase page count.

- c) A suggested font setting is provided in the appendix.

- **Presentation** must be entirely in English, including both spoken language and presentation materials. While presentation materials do not have to be PPT, it is recommended not to spend excessive effort on styling; the goal is to clearly present the report content.

- **Random grouping.** Three-person groups receive no score compensation compared to four-person groups; four-person groups will present first, three-person groups later.

- **Optional tasks** have some influence on the score, at least in the areas of R4 engagement and R6 use of mathematics. However, do not pursue too many optional tasks. It is suggested that each group either pick one optional task or none. The aim is to guide you to explore beyond the given data based on your interests and abilities.

- **References** in the report should be as standardised as possible, using the Chicago format: <https://lib.tsinghua.edu.cn/info/1147/3359.htm>. Students are encouraged to follow this standard. In any case, Baidu Baike and Wikipedia are not acceptable as references; legitimate media articles are acceptable.

- **Remove of records.** Based on the original dataset, we have removed borrowing records of A-Level and OSSD students, keeping only IB students' borrowing information. This is to reduce data size and ensure fair comparison across grades (Grade 22 has no A-Level).

- **Choice of dataset.** The dataset is available in two versions: full dataset and a 40% sampled dataset.

The full dataset contains approximately 2,000 borrow or renew records, which can be analysed with the help of AI, programming tools, etc., which should be stated in the

report.

The sampled dataset contains about 800 borrow or renew records, which can be analysed manually without heavy use of AI or programming. You can always use the help of Excel or other softwares.

Please clearly indicate which dataset you choose at the beginning of the report. The choice does not affect the marking of the monthly assessment, but reports using the full dataset will be given priority for recommendation to the librarians.

Regarding AI assistance

- a) Use of AI is not prohibited.
- b) However, you must not copy-paste AI-generated sentences directly into the report.
- c) Briefly describe your use of AI in the appendix.
- d) In the appendix, attach all (helpful to the report) prompts given to AI and AI's responses.

The intention is to help you master AI as a tool, not to rely on it.

References

国家图书馆《中国图书馆分类法》编辑委员会. 2010. 《中国图书馆分类法》. 第5版. 北京: 国家图书馆出版社.

Dewey, Melvil, Joan S. Mitchell, Julianne Beall, Rebecca Green, Giles Martin, and Michael Panzer. Dewey Decimal Classification and Relative Index. Edition 23. Dublin, Ohio: OCLC Online Computer Library Center, Inc., 2011.

Appendix

A suggested font setting:

For English academic papers, font, size, and line spacing requirements generally follow international academic standards (e.g., APA, MLA, Chicago) or specific journal/school format guidelines. Common requirements are as follows:

1. General standards (e.g., APA 7th/MLA)**- Font:**

- Recommended: Times New Roman or Arial, etc.
- Alternatives: Calibri, Georgia, or 12-point Courier New.

- Size:

- Body text: 12pt
- Headings: 14–16pt bold (e.g., level-1 heading)
- Header/Footer: 10–12pt

- Line spacing:

- Body text: 1.5 line spacing
- Figure captions, footnotes: Single line spacing

- Paragraph spacing:

- If first-line indent (0.5 inch / 1.27 cm) is used, no blank line between paragraphs; if no indent, a blank line between paragraphs.

2. Other details

- Margins: Usually 1 inch (2.54 cm) on all sides.
- Alignment: Body text left-aligned, avoid justified.
- Heading format:
 - APA: hierarchical headings with different sizes and bold (e.g., level-1 heading 14pt bold, centered).
 - MLA: headings centered, no extra formatting.

3. Notes

- Figures/tables and references: Usually same font size as body text, but line spacing

may be adjusted to single.

- Footnotes/endnotes: Usually 10pt, single line spacing.

(Adapted from DeepSeek's answer to "What are the font, size, and line spacing requirements for English papers?")

Use of AI:

1. This article was written using Deepseek, based on the question: What are the font, size, and line spacing requirements for English papers? The answer is shown above.
2. Google Translate was also used to help refine the text.